VirtualBox and Kali Linux Installation (Revision Notes)

Module 01 – VirtualBox and Kali Linux Installation

1. Virtual Machine (VM)

- A Virtual Machine is software that allows multiple operating systems to run on a single computer.
- Each virtual machine works independently with its own operating system and resources.

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2. VirtualBox

- Oracle VM VirtualBox is a free and open-source virtualization software.
- It allows users to create and manage virtual machines.
- Multiple operating systems can run simultaneously on one physical computer.

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3. Why Use VirtualBox?

- Test different operating systems safely.
- Practice cybersecurity without affecting the host system.
- Create isolated environments for learning and experimentation.

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4. Kali Linux

- Kali Linux is a Debian-based Linux distribution.
- Developed and maintained by **Offensive Security**.
- Designed for penetration testing, digital forensics, and security research.

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5. Downloading VirtualBox

- Download the latest version compatible with your operating system.
- Install VirtualBox before creating a virtual machine.

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6. Downloading Kali Linux

- Download the official Kali Linux VirtualBox image.
- Use the pre-built VirtualBox image for quick setup.

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7. Importing the Virtual Machine

- Open VirtualBox.
- Import or open the downloaded Kali Linux image.
- Verify the virtual machine settings before starting it.

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8. Virtual Machine Settings

Configure the following resources before starting the VM:

- RAM (Memory)
- Processor (CPU)
- Storage
- Network Adapter
- Display Settings

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9. Starting Kali Linux

- Start the virtual machine from VirtualBox.
- Wait for Kali Linux to boot successfully.
- Log in using the provided credentials if required.

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10. Network Configuration

- Configure the network adapter according to the requirement.
- Ensure internet connectivity if updates or downloads are needed.

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11. Shared Clipboard

- Enable Shared Clipboard for easy copy and paste between the host and virtual machine.
- Bi-directional mode allows copying in both directions.

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12. Shared Folder

- Shared Folders allow files to be exchanged between Windows (host) and Kali Linux (guest).
- Configure the folder in VirtualBox settings before starting the VM.

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13. Guest Additions

- VirtualBox Guest Additions improve integration between the host and guest operating systems.
- Features include:
  - Better display resolution
  - Clipboard sharing
  - Drag-and-drop support
  - Shared folders

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14. Best Practices

- Allocate sufficient RAM and CPU resources.
- Keep VirtualBox and Kali Linux updated.
- Take snapshots before making major system changes.
- Use the VM only for authorized learning and practice.

Key Takeaways

- Learned how virtualization works.
- Understood the purpose of VirtualBox.
- Learned how to install and configure Kali Linux.
- Explored VirtualBox features such as Shared Clipboard and Shared Folders.
- Understood the importance of Guest Additions and proper VM configuration.